

# Experiment 15

Design a C program to organise the file using a two level directory structure.

## C Program

```
#include <stdio.h>
#include <string.h>

#define MAX_USERS 10
#define MAX_FILES 20
#define NAME 30

char users[MAX_USERS][NAME];
char files[MAX_USERS][MAX_FILES][NAME];
int fileCount[MAX_USERS];
int userCount = 0;

int findUser(char name[]) {
    for (int i = 0; i < userCount; i++)
        if (strcmp(users[i], name) == 0) return i;
    return -1;
}

int main() {
    int choice, u;
    char user[NAME], file[NAME];

    while (1) {
        printf("\n1. Create User\n2. Create File\n3. Search File\n4. Display Directory\n5. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        if (choice == 1) {
            printf("Enter user name: ");
            scanf("%29s", user);
            if (userCount < MAX_USERS && findUser(user) == -1) {
                strcpy(users[userCount], user);
                fileCount[userCount] = 0;
                userCount++;
                printf("User directory created.\n");
            } else printf("User already exists or directory is full.\n");
        } else if (choice == 2) {
            printf("Enter user name: ");
            scanf("%29s", user);
            u = findUser(user);
            if (u == -1) {
                printf("User not found.\n");
                continue;
            }
            printf("Enter file name: ");
            scanf("%29s", file);
            if (fileCount[u] < MAX_FILES) {
                strcpy(files[u][fileCount[u]++], file);
                printf("File created in %s directory.\n", user);
            }
        } else if (choice == 3) {
            printf("Enter user name: ");
            scanf("%29s", user);
            u = findUser(user);
            if (u == -1) {
                printf("User not found.\n");
                continue;
            }
            printf("Enter file name: ");
            scanf("%29s", file);

            int found = 0;
            for (int i = 0; i < fileCount[u]; i++)
                if (strcmp(files[u][i], file) == 0) found = 1;

            printf(found ? "File found.\n" : "File not found.\n");
        } else if (choice == 4) {
            for (int i = 0; i < userCount; i++) {
```

```
        printf("%s/: ", users[i]);
        for (int j = 0; j < fileCount[i]; j++)
            printf("%s ", files[i][j]);
        printf("\n");
    }

    } else if (choice == 5) break;
    else printf("Invalid choice.\n");
}

return 0;
}
```

## Sample Output

```
1. Create User
2. Create File
3. Search File
4. Display Directory
5. Exit
Enter choice: 1
Enter user name: user1
User directory created.
```

```
Enter choice: 2
Enter user name: user1
Enter file name: notes.txt
File created in user1 directory.
```

```
Enter choice: 4
user1/: notes.txt
```

## Result

The two-level directory structure was successfully simulated by maintaining separate user directories and their files.